

Step 1: Requirements Analysis

University Intelligent Assistant

Project: An AI-Powered Intelligent Assistant Using Dify

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1. Purpose of the System

The proposed system is a University Intelligent Assistant that uses artificial intelligence to help university students obtain clear and organized information through a conversational interface. The assistant will receive students' natural-language questions and provide relevant responses based on the configured instructions and available university information.

2. Target Users

The primary target users are university students who need quick access to information related to their academic study and university services.

3. Main User Requests

- Questions about university courses and academic information.
- Questions about registration and academic procedures.
- Questions about class schedules and related academic arrangements.
- Questions about university regulations and policies.
- Questions about available university services.

4. Required Information

The assistant may use relevant university documents, regulations, academic information, schedules, and other approved reference materials as a knowledge source. The exact documents and data sources will be identified during the system design and Dify configuration stages.

5. Functional Requirements

- 1 The system shall provide a conversational interface for students.
- 2 The system shall receive and interpret natural-language questions.
- 3 The system shall generate responses according to configured instructions.
- 4 The system shall use available reference information when required.
- 5 The system shall return responses in a clear and organized format.
- 6 The system shall support testing using representative student questions.

6. Non-Functional Requirements

- Usability: the interface should be simple and easy for students to use.
- Clarity: responses should be understandable and organized.

- Consistency: similar questions should receive reasonably consistent responses.
- Relevance: responses should address the user's question.
- Accuracy: responses should be evaluated against available reference information.

7. System Limitations and Scope

The assistant will operate within the defined university-information scope. It will not be treated as a system capable of answering every possible question. Response quality may depend on the selected language model, prompts, and the quality and availability of reference information.

8. Initial Test Questions

- 1 What courses are available for my program?
- 2 How can I register for a course?
- 3 Where can I find the academic schedule?
- 4 What are the university rules for registration?
- 5 What student services are available?

9. Expected Outcome of Step 1

This requirements analysis defines the initial purpose, users, request categories, information needs, functional requirements, limitations, and preliminary test cases of the University Intelligent Assistant. These requirements will be used as the basis for the next stage: Application Design.

Next Step: Application Design — defining the conversational structure, system instructions, prompt strategy, workflow steps, and required knowledge sources.